Weekly Progress Log – Week 6

Project Title: Data Science with Python
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Week: 6

Objectives

- Understand confidence intervals, margin of error, and what a p-value means.
- Learn how to state null (H_0) and alternative (H_1) hypotheses and interpret statistical significance.
- Study and apply a t-test and a chi-square test using sample data.
- Understand the basic process of A/B testing and its importance for business decisions.
- Apply at least one hypothesis test to real data in a Jupyter Notebook and explain the result in plain language.

Work Completed

- Prepared study notes on confidence intervals and margin of error, including examples using average test scores and customer satisfaction percentages.
- Studied the purpose of null and alternative hypotheses, p-values, the 0.05 significance level, and the difference between rejecting and not rejecting H_0.
- Prepared notes explaining t-tests, chi-square tests, and the use of each test for numerical or categorical data.
- Studied A/B testing using an online-store button example, including random assignment, conversion rates, and business benefits.
- Used the real Tips dataset (244 restaurant records and seven variables), and checked its structure and missing values using Pandas.
- Performed a Welch independent samples t-test comparing lunch (68 bills) and dinner (176 bills): mean bills \$17.17 and \$20.80; $p = 0.0022$.
- Calculated a 95% confidence interval for lunch minus dinner bill averages, approximately -\$5.93 to -\$1.33.
- Performed a chi-square test of independence for smoking status and meal time: $\chi^2 \approx 0.736$ and $p = 0.3910$.
- Created box plots, grouped bar charts, an observed-frequency table, and an expected-frequency table to explain the results.
- Completed a Jupyter Notebook with test questions, H_0/H_1, Python code, p-values, decisions, and plain-language conclusions.

Challenges Faced

- Distinguishing H_0 from H_1 and understanding why their wording differs between a t-test and a chi-square test.

- Understanding what a p-value represents and when $p < 0.05$ means a statistically significant result.
- Choosing a suitable test for averages versus relationships between categories.
- Interpreting a negative confidence interval for the difference between lunch and dinner averages.
- Separating statistical significance from practical importance or a claim of cause and effect.
- Understanding why the observed restaurant comparisons are not a randomized A/B test.

Solutions Implemented

- Wrote the research question and H_0/H_1 before each test and compared each calculated p-value with $\alpha = 0.05$.
- Applied Welch's t-test to numerical total bills and a chi-square independence test to meal time and smoking-status categories.
- Used observed and expected frequency tables and graphs to make the chi-square test easier to interpret.
- Calculated and interpreted the 95% confidence interval alongside the t-test p-value.
- Explained each result in everyday language without claiming that the data proves cause and effect.
- Kept the real CSV dataset and statistical notebook as separate files; reviewed the need to use a portable relative data path when sharing the notebook.
- Prepared supporting study documents on confidence intervals, margin of error, t-tests, chi-square tests, and A/B testing.

Individual Tasks

Responsibility	Description of Contribution	Remark
Confidence Intervals & MOE	Studied interval estimation, margin of error, and worked examples.	Prepared confidence-interval study notes.
Hypothesis Testing Basics	Reviewed H_0, H_1, p-values, and the 0.05 decision rule.	Improved understanding of test interpretation.
Real Dataset Preparation	Loaded and explored the 244-row Tips dataset using Pandas.	Confirmed dataset structure and missing values.
T-Test Application	Compared mean lunch and dinner bills using Welch's t-test and a 95% interval.	$p = 0.0022$; significant difference.
Chi-Square Application	Tested smoking status versus meal time using a contingency table.	$p = 0.3910$; insufficient evidence of association.

A/B Testing Study	Reviewed random assignment, conversion rates, and business use cases.	Prepared A/B testing study notes.
Visualization & Conclusions	Created comparison charts and wrote plain-language interpretations.	Results summarized clearly in the notebook.
Documentation	Prepared Week 6 study PDFs and the hypothesis-testing notebook.	Deliverables organized for review.